Unit-I

Introduction to Web Technologies

HTML: Getting started with HTML, Why HTML, Tags and Elements, Attributes, Properties, Headings list, Links, Tables, Images, HTML Form, Media (Audio, Video), Semantic HTML5 Elements.
CSS: Why CSS, Types of CSS, How to use CSS, Properties, Classes, Child-Class (Nested CSS), Colors, Text, Background, Border, Margin, Padding, Positioning (flex, grid, inline, block), Animation, Transition.
BOOTSTRAP: Why Bootstrap, CSS over Bootstrap, How to Use Bootstrap, Bootstrap Grid System, Bootstrap Responsive, Bootstrap Classes, Bootstrap Components (i.e., Button, Table, List, etc.),Bootstrap as a Cross Platform.
W3C: What is W3C , How W3C handles/Supports Web Technologies.

What is HTML?

- HTML stands for Hyper Text Markup Language.
- HTML is the standard markup language for creating Web pages.
- HTML describes the structure of a Web page.
- HTML consists of a series of elements.
- HTML elements tell the browser how to display the content.

Why Learn HTML?

Here are 5 common reasons to learn HTML:

1. **Build Websites**: HTML is the basic building block for creating any website. Learning HTML can help you pursue a career in web development.
2. **Customize Content**: Allows you to edit or tweak web pages, emails, or templates to fit your needs.
3. **Understand how the web works**: Helps you grasp how the internet works and how web pages are structured.
4. **Employment Opportunities**: According to Bureau of Labor Statistics projects that employment for web developers will grow 16% between 2022-2032, which is much faster than the average across all occupations.
5. **Learn Easily**: HTML is beginner-friendly, making it a great first step into the world of coding and technology

HTML TAG

- An HTML tag is a keyword that instructs a web browser on how to display and format a web page.

- HTML tags are enclosed in angle brackets (< >) and are made up of elements and attributes.
- HTML tags are not case sensitive, but it is recommended to write them in lowercase.
- The content should be enclosed b/w tags.
- The container and contain together called elements.
- Ex
    1. <html>: The opening tag for the root of an HTML document. The closing tag is </html>.
    2. <main>: A structural element that represents the main content of a document.
    3. <aside>: Represents a portion of a document that is indirectly related to the main content, such as a sidebar or call-out box.
    4. <footer>: Represents a footer for the nearest ancestor sectioning content or sectioning root element.
    5. <header>: Represents introductory content, such as a logo, search form, or author name.
    6. <img>: Used to embed an image in an HTML page.
    7.
: A self-closing tag that creates a line break in the text.

HTML Elements

- An HTML element is a component of an HTML document that defines how a web browser should interpret a part of the document.
- HTML elements are the building blocks of web pages, and each element represents a different part of the page's content.

What do HTML elements do?

- Add formatting: HTML elements can make text bold, italicized, or underlined, or organize it into paragraphs, lists, and tables.
- Add semantic meaning: HTML elements can indicate important or emphasized text, or create a hyperlink.
- Embed content: HTML elements can embed images, videos, or other elements.

Example

<p>Hello World </p>

- <a>... Creates a hyperlink
- ... Emphasizes text by italicizing it
- ... Emphasizes text by making it bold
-
 Creates a line break
- ... Creates an unordered list
- ... Creates an ordered list

Attributes

- Attributes are used to specify the alternate meaning of tag, it can be appears b/w an opening tag and right angler bracket (>).
- These are in keyword form, followed by equal sign and attribute value.
- Attribute names should be in lower case and value should enclosed between (“ “) double quotes.

Ex

< p align= “center” > Hello World </p>

Comments

- Comments in a program increase the readability of a program.
- Syntax is < ! --- -- >
- Generally comments are used as follows
  - Name of the Application
  - Description of code file
  - Name of Author

Structure of HTML Documents

```
    <html>
  <head>
      <title> First Page </title>
   </head>
    <body>
      <h1>Welcome to HTML</h1>
      <p>Description-------- </p>
             </body>
      </html>
```

- HTML document can save with the extension of **.html or .htm.**
- Every HTML documents surrounded by

<html>- - - - - - - - - - - - - - -- - - - - </html >

- Head section contains control information used by browser and server.
- Head section contains the contains of documents that display on screen and tags
- Every time HTML pages having one title which is displayed at the top of browser window.
- Some of the generated tags are:
- **Paragraph Tag**

< p[align=”left ”| “right” | ”center”] >- - - - -</p>

- **Heading Tag**

<h1 [align=”left ”| “right” | ”center”] >- - - - -</h1>

<h2>- - - - - - - - - - -</h2>

- **<h5> & <h6>** are used for smaller heading.
- **<br** /> -breaking the sentence from one line to another line,slash indicates that the tag is both opening and closing.
- < **b**> **-** bold the text

- - - - - - - -

- <**i**> - Italic text

<i>- - - - - - - -</i>

- **<strong** > - Highlighting the text or stress the text

<strong>- - - - - - - -</ strong >

- <**strike** > **-** Striking the text

<strike>- - - - - - - -</ strike>

- <**sup** > **-** Superscript the text

<sup>- - - - - - - -</ sup>

- <**sub** >**-** Subscript the text

<sub>- - - - - - - -</ sub>

- <**big**> **-** Large the text

<big>- - - - - - - -</ big>

- <**small** > - Smaller the text

<small>- - - - - - - -</ small>

- <**tt** >**-** typewritten the text or Monospace font style

<tt>- - - - - - - -</ tt>

Character Entries:

Character	Entity	Meaning
(space)		non-breaking space
<	<	less than
>	>	greater than
&	&	ampersand
"	"	double quotation mark
'	'	single quotation mark
©	©	copyright
®	®	trademark

Example

```
<!DOCTYPE html>
<html>
<body>
<h1>HTML Entity Example</h1>
<h2>The less-than sign: &lt;</h2>
</body>
</html>
```

Output

HTML Entity Example

The less-than sign: <

Hyperlinks

- The <a> tag defines a hyperlink, which is used to link from one page to another.
- The most important attribute of the <a> element is the href attribute, which indicates the link's destination.

By default, links will appear as follows in all browsers:

- An unvisited link is underlined and blue.
- A visited link is underlined and purple.
- An active link is underlined and red.

- It has the following syntax:

link text

Example

```
<!DOCTYPE html>
<html>
<body>

<h1>HTML Links</h1>
<p><a href="https://www.w3schools.com/">Visit W3Schools.com!</a></p>
</body>
```

</html>

Output

HTML Links

[Visit W3Schools.com!]

List Tag

- Lists are used for effective way of structuring the web pages content.
- It provides straight forward index in website.
- There are 3 types of lists
    1. Ordered List
    2. Unordered List
    3. Definition List

1 Ordered List (Numbered)

These lists are used when the order of the items is important. Each item in an ordered list is typically marked with numbers or letters.

An ordered list starts with the <ol> tag. Each list item starts with the <li> tag.

```
<ol>
   <li>Coffee</li>
   <li>Tea</li>
   <li>Milk</li>
 </ol>
```

Example

```
<!DOCTYPE html>

<html>

<body>

      <h2>An ordered HTML list</h2>

     <ol>

  <li>Coffee</li>

 <li>Tea</li>

 <li>Milk</li>

    </ol>

</body>

</html>
```

Output

An ordered HTML list

1. Coffee
2. Tea
3. Milk

2 Unordered List (Bulletted)

- An unordered list starts with the <ul> tag. Each list item starts with the <li> tag.
- The list items will be marked with bullets (small black circles) by default:

```
<ul>
   <li>Coffee</li>
   <li>Tea</li>
   <li>Milk</li>
 </ul>
```

Example

```
<!DOCTYPE html>

<html>

<body>

   <h2>An unordered HTML list</h2>

      <ul>

 <li>Coffee</li>

 <li>Tea</li>

 <li>Milk</li>

     </ul>

</body>

</html>
```

Output

An unordered HTML list

- Coffee
- Tea
- Milk

3 Definition List

- A definition list is a list of terms, with a description of each term.
- The <dl> tag defines the description list, the <dt> tag defines the term (name), and the <dd> tag describes each term.

```
<dl>
  <dt>Coffee</dt>
  <dd>- black hot drink</dd>
  <dt>Milk</dt>
  <dd>- white cold drink</dd>
 </dl>
```

Example

```
<!DOCTYPE html>

<html>

<body>

<h2>A Description List</h2>

     <dl>

          <dt>Coffee</dt>

          <dd>- black hot drink</dd>

       <dt>Milk</dt>

         <dd>- white cold drink</dd>

     </dl>

</body>

</html>
```

Output

A Description List

-Coffee

- black hot drink

-Milk

-white cold drink

Tables

- Table are providing highly readable way of presenting many type of information
- HTML tables allow web developers to arrange data into rows and columns in which intersection of rows and column is called cell.
- Each table cell is defined by a <td> and a </td> tag.

td stands for table data.

- Each table row starts with a <tr> and ends with a </tr> tag

tr stands for table row.

th stands for table header.

Example

```
<!DOCTYPE html>
<html>
<body>
  <table>
    <tr>
      <th>Firstname</th>
      <th>Lastname</th>
      <th>Age</th>
    </tr>
    <tr>
      <td>Priya</td>
      <td>Sharma</td>
      <td>24</td>
    </tr>
    <tr>
      <td>Arun</td>
      <td>Singh</td>
      <td>32</td>
    </tr>
    <tr>
      <td>Sam</td>
      <td>Watson</td>
      <td>41</td>
    </tr>
  </table>
</body>
</html>
```

Output

Firstname	Lastname	Age
Priya	Sharma	24
Arun	Singh	32
Sam	Watson	41

HTML Image

- The inclusion of image in to documents will display image on web sites
- The <img> tag is used to embed an image in an HTML page.
- The <img> tag has two required attributes:

1. src - Specifies the path to the image
2. alt - Specifies an alternate text for the image, if the image for some reason cannot be displayed

- GIF- Graphic Intercharge Format
- BMP-Bit Map Image
- PNG-Portable Network Graphics
- TIFF-Tag Index File Format

Syntax:

```
<img src="img_girl.jpeg" alt="Girl in jacket" width="500" height="600">
```

HTML FORM

- An HTML form is used to collect user input. The user input is most often sent to a server for processing.
- The HTML <form> element is used to create an HTML form for user input:

```
<form>
  action=”URL”
Method=”GET/POST”
   .
   .
   .
   .
   .
  </form>
```

action attribute - action should be perform or target page when form is to be submitted.

Method attribute-

GET-User information is visible on address bar of web page.

POST- It will hide the information of web page.

Controls in Form

Generally in any applications control are like text, radio button, checkbox select from list, drop down list, submit and reset button, text area field.

I. **text:-** It uses single line of text ,every attribute is separated by name attribute.

<input type=”text” name=”string” >

II. **password:-** It is similar to text but contain is not display on the screen

<input type=”password” name=”string”/ >

III. **checkbox:-** It simply provided multiple options.

<input type=”checkbox”[checked] name=”string”/ >

- Checked attribute is optional if we write checked then first or corresponding value is default selected.

IV. **Radio Button:-** They are always group together with same name but different values. From multiple options we are selecting one option.

<input type="radio" id="html" name="fav_language" value="HTML">

<input type="radio" id="css" name="fav_language" value="CSS">

V. **Text area:-** For longer than single sentence we used it.

<textarea cols=”n” rows=”n” name=”string”/>

VI. **Select:-** It will create button which display the value can be select
- Dropdown box which provides options list.

<select name=”dropdown”>
<option value=”Maths” selected>Maths </option>
<option value=”English” >English </option>
</select>

Button Controls

a. **submit:** It will used for sending or storing user information.

<input type="submit" value="submit" name="string"/>

b. **Reset:** It is used for clear the information in the form.

<input type= ”reset” value=”clear” name=”string”/ >

c. Button: It is used for clicking purposes

<input type=”Button” value=”ok” name=”string”/ >

d. Image Button: It is used for clicking purposes.

<input type=”images ” name=”string” src=URL />

e. File upload box:- It allow users to upload a file to the website.

<input type=”file ” name=”string” accept=”image/*” />

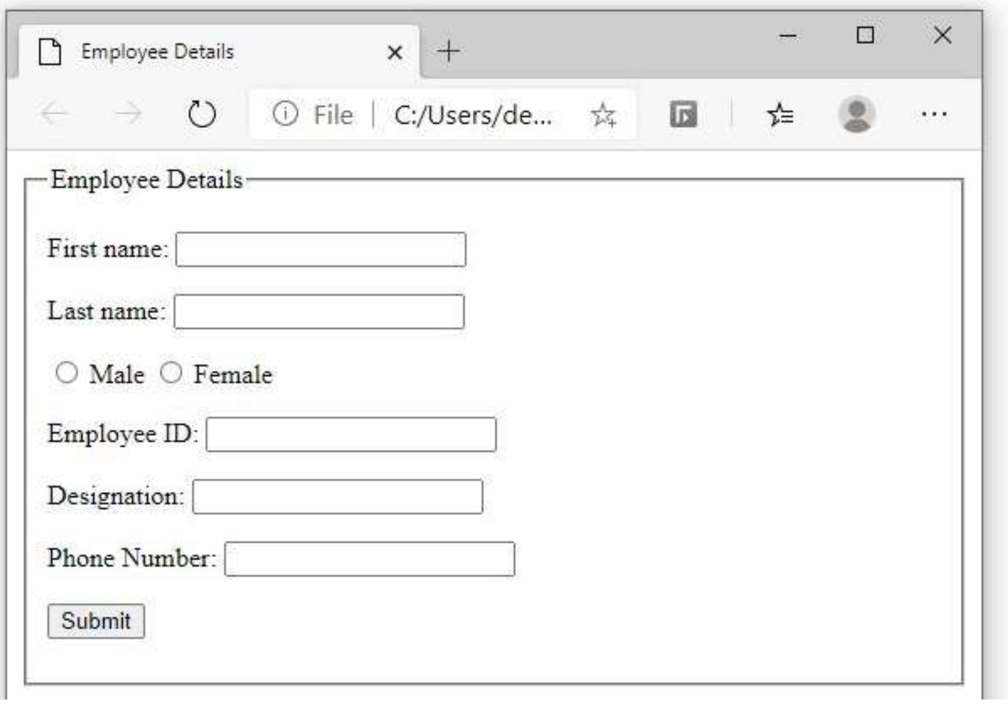
- accept attribute indicates that type of file we are uploading.

Example

```
<!DOCTYPE html>
<html lang="en">
<head>
<title>Employee Details</title>
</head>
<body>
  <form>
    <fieldset>
    <legend>Employee Details</legend>
     <p>
         First name: <input type = "text" name = "fname" />
  </p>
   <p>
         Last name: <input type = "text" name = "lname" />
  </p>
   <p>
         <input type = "radio" name = "Gender" value = "Male"> Male
         <input type = "radio" name = "Gender" value = "Female"> Female
</p>
<p>
Employee ID: <input type = "text" name = "ID" />
</p>
<p>
Designation: <input type = "text" name = "ID" />
</p>
<p>
Phone Number: <input type = "text" name = "phone" />
</p>
<p>
      <input type = "submit" name = "submit" value = "Submit" />
```

</p>
</fieldset>
</form>
</body>
</html>

Output



FRAMES

- For displaying more than one document at a time we are using FRAMES.
- Here window is divided into rectangle area each of which is called frame.
- Each frame is capable of displaying its own document.
- Generally frames are used the table of content will display in one page and part of the main document display in another frame.

< frameset cols=”50 % * ” rows=”50 % *”

.

. **<! ---Frame Details --->**

.

</frameset>

- No of frames and its layout of a web browser will decide by rows and cols attribute.
- frames :

  - the content of the frames is specified

  - It will display the content of frameset

< frame name= ”string” src= ”filename”

Scrolling= ”yes” | ”auto” | ”no”

frameborder= ”0” | ”1”/ >

Frames Name & target attribute

```
<!DOCTYPE html>
<html>
<head>
<title>HTML Target Frames</title>
</head>
 <frameset cols="200, *">
  <frame src="/html/menu.htm" name="menu_page"/>
<frame src="/html/main.htm" name="main_page" /> <noframes>
 <body> Your browser does not support frames. </body>
</noframes>
 </frameset>
</html>
```

CSS

- CSS stand for Cascading Style Sheet.
- HTML is concentrate on the content rather than the presentation of content.
- css is markup language used in web documents for presentation purpose.
- There are 3 types of CSS

1. Inline CSS
2. Internal or Embedded CSS
3. External CSS

1.Inline CSS

- An inline CSS is used to apply a unique style to a single HTML element.

- An inline CSS uses the style attribute of an HTML element.

Syntax:

<tag-name style=”property:value;”>

Ex:
```
<p style="color:#009900;
    font-size:50px;
    font-style:italic;
    text-align:center;">
  </p>
```

Example

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Document</title>
</head>
<body>
  <h1 style="color:blue;">A Blue Heading</h1>
<p style="color:red;">A red paragraph.</p>
</body>
</html>
```

<u>Output</u>

A Blue Heading

A red paragraph.

Drawbacks:

- By using this style sheet we can apply uniform style on tags for whole documents.
- Content of web pages is mixed with presentation.

2. <u>Internal or Embedded CSS</u>

- It will appear in the head section and in the body section newly defined selector tags are used with actual content.

Syntax: style specification format

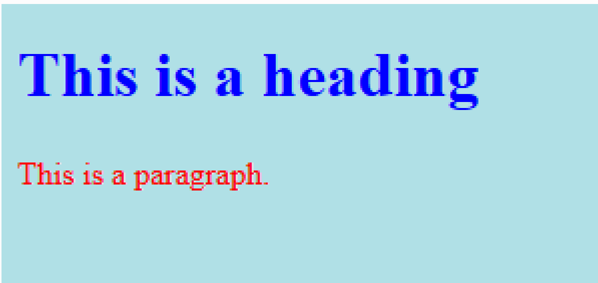
```
<style type=" text / css ">
     Selector{
              property : value;
             property: value;
     .
     .
             }
     </style>
```

Example

```
<!DOCTYPE html>
<html lang="en">
```

```
<head>
 <meta charset="UTF-8">
 <meta name="viewport" content="width=device-width, initial-scale=1.0">
 <style>
  body {background-color: powderblue;}
  h1  {color: blue;}
  p   {color: red;}
  </style>
 <title>Document</title>
</head>
<body>
 <h1>This is a heading</h1>
<p>This is a paragraph.</p>
</body>
</html>
```

Output



Drawbacks:

- When we want to apply the style to more than one documents at a time it is not possible.

3.External CSS

- External CSS contains separate CSS files that contain only style properties with the help of tag attributes.
- CSS property is written in a separate file with a .css extension and should be linked to the HTML document using a link tag.

Syntax:

<link rel=”stylesheet” type =”text / css” href=”.css *”/>

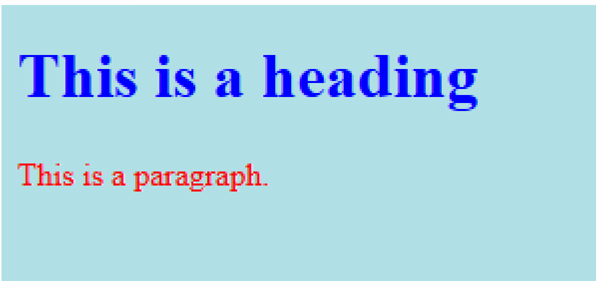
Example

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <link rel="stylesheet" href="style.css">
  <title>External Styele Sheet</title>
</head>
<body>
  <h1>This is a heading</h1>
<p>This is a paragraph.</p>
</body>
</html>
```

styles.css

```
body {
    background-color: powderblue;
  }
  h1 {
    color: blue;
  }
  p {
    color: red;
  }
```

Output



Some of CSS Categories properties & values

<u>fonts</u>
font-family : family name;
font-style : Normal | italic | obliques ;
font-weight : Normal | bold | bolder | lighter ;
font-sizes : small | smaller |medium |large |larger ;

<u>Backgrounds and colors</u>
color : values ;
background – color : values ;
background – images : URL ;
background - repeat : repeat -x | no -repeat ;
background – position : center | top | bottom |left | right ;

<u>Text</u>
text-decoration : none | underline | overline | blink | line-through ;
text-transformation : uppercase | lowercase | capitalize | none ;
text-align : left | right | center | justify ;
text-height : length | percentage ;
text-spacing : length | percentage ;
margin- right : 200 x;
margin -left : 300 x ;

<u>Link</u>
a: link { color : green ; font -size ; 18 px ;}
a : hover { color : orange ;}
a : visited { color : blue ;}
a : active { color : black ;}

<u>List</u>
List-style-type : roman | lower-greek | upper -latin ;
List-style-image : url | (“ new.git ”);

<u>Border</u>
border-style : dashed | solid | double | dotted ;
border-color : color ;
border-width : top | bottom | left | right ;

<u>Table</u>
width : length | percentage ;
border : solid | dotted ;
vertical -align : bottom | center ;

Example:-

```
<!DOCTYPE html>
<html>
<head>
<style>
h1,p
   {
      text-align : center ;
         color : red ;
   }
h2
   {
      background-color : green ;
   }



body
   {
      background -color : lightblue ;
   }
</style>
</head>
<body>
<h1>This is a heading</h1>
<p>This is a paragraph</p>
</body>
</html>
```

BootStrap

- Bootstrap is a popular front-end framework for building responsive and mobile-first websites. It provides pre-designed CSS, JavaScript components, and utility classes to quickly create modern and consistent user interfaces.
- It Includes pre-built responsive grid systems for mobile-first design.
- Offers a wide range of UI components like buttons, modals, and navbars
- Provides built-in support for responsive typography, spacing, and utilities.
- Extensively customizable via Sass variables and Bootstrap's configuration.

CSS over Bootstrap

Bootstrap	**CSS**
Bootstrap is a front-end framework designed to help developers quickly create mobile-first, responsive websites.	CSS (Cascading Style Sheets) is a stylesheet language used to describe the presentation of a document written in HTML or XML.
Used for developing responsive and mobile-first websites with a set of predefined CSS and JavaScript components.	It is used to style and layout web pages, for example, changing the font, colour, size, and spacing of content, splitting it into multiple columns, or adding animations and other decorative features.
Has a steeper learning curve for absolute beginners, as it requires basic knowledge of CSS (and optionally HTML & JavaScript) to use the framework effectively.	CSS is fundamental to web development, and understanding it is crucial for styling web pages. It can be simpler to learn from scratch, but mastering it takes time.
Offers extensive customization options through predefined classes and components, but extensive customizations can increase complexity.	Allows for complete customization from the ground up, providing the flexibility to create unique designs without predefined components.
Enables rapid development of layouts and components, making it ideal for projects with tight deadlines.	Developing unique designs from scratch can be time-consuming compared to using a framework like Bootstrap.
Has a large community and extensive documentation, making it easier to find solutions to problems or learn from existing templates and themes	While CSS itself has widespread support, specific design solutions require individual research or consultation of community forums and documentation
Including the full Bootstrap framework can lead to larger file sizes, potentially affecting load times for simple websites.	CSS files are typically smaller than including a full framework, which can lead to faster page load times, especially for simple websites.

- To link HTML page to Bootstrap following link are used:

  1. **<u>Include via CDN</u>**

```
<link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/bootstrap.min.css"
  rel="stylesheet" integrity="sha384-
  QWTKZyjpPEjISv5WaRU9OFeRpok6YctnYmDr5pNlyT2bRjXh0JMhjY6hW+AL
  EwIH" crossorigin="anonymous">
```

  2. **<u>Install via package manager</u>**

```
npm install bootstrap@5.3.3

gem install bootstrap -v 5.3.3
```

Example

```
<!DOCTYPE html>
<html lang="en">
<head>
 <meta charset="UTF-8">
 <meta name="viewport" content="width=device-width, initial-scale=1.0">
 <title>Bootstrap</title>
<link
href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/bootstrap.min.css"rel="st
ylesheet"  integrity="sha384QWTKZyjpPEjISv5WaRU9OFeRpok6YctnYmDr5pNly
T2bRjXh0JMhjY6hW+ALEwIH" crossorigin="anonymous">
</head>
<body>
   <a href="" class="btn btn-primary">click here</a>
</body>
   </html>
```

<u>Bootstrap Grid System</u>

The Bootstrap Grid System is a layout system used to create responsive web pages using rows and columns.

It divides the page into 12 equal columns and uses flexbox to arrange content.

Why Grid System?

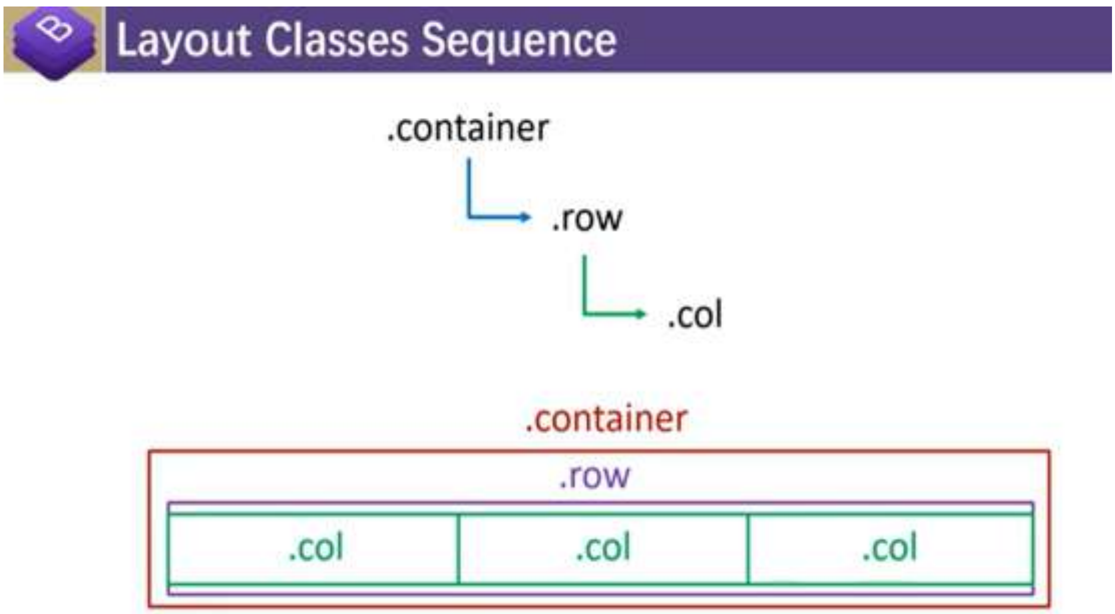
It helps you:
✔ Arrange content easily
✔ Create responsive layouts

✓ Align elements properly
✓ Work on all screen sizes

Basic Structure

Bootstrap grid always follows:

```
<div class="container">
  <div class="row">
    <div class="col">Content</div>
  </div>
</div>
```



1.container

- ➢ Wraps the content
- ➢ Adds margin & padding
- ➢ Centers the layout
- ➢ Syntax

```
<div class="container">
```

Example

```
<!DOCTYPE html>
<html lang="en">
<head>
```

```
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0" >
    <title>First Bootstrap Program</title>
    <link
   href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.8/dist/css/bootstrap.min.css"
   rel="stylesheet" integrity="sha384-
   sRIl4kxILFvY47J16cr9ZwB07vP4J8+LH7qKQnuqkuIAvNWLzeN8tE5YBujZqJ
   LB" crossorigin="anonymous">
    <link rel ="stylesheet" href="style.css">
</head>
<body>
    <div class="container"> Hello</div>
</body>
</html>
```

style.css

```
.container{
    border:8px solid black;
}
```

2) .row

- Creates horizontal group
- Uses flexbox
- Holds columns
- syntax

<div class="row">

Example

```
<!DOCTYPE html>

<html lang="en">

<head>

   <meta charset="UTF-8">

   <meta name="viewport" content="width=device-width, initial-scale=1.0" >

   <title>First Bootstrap Program</title>

   <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.8/dist/css/bootstrap.min.css"
  rel="stylesheet" integrity="sha384
  sRIl4kxILFvY47J16cr9ZwB07vP4J8+LH7qKQnuqkuIAvNWLzeN8tE5YBujZqJLB
  " crossorigin="anonymous">

   <link rel ="stylesheet" href="style.css">
</head>
<body>
   <div class="container">
```

```
   <div class="row">Row1 </div>
    <div class="row">Row2 </div>
     <div class="row">Row3 </div>
   </div>
</body>
</html>
```

Style.css

```
.container{
   border:8px solid black;
}
.row{

   border:5px solid red;
}
```

3).col

- Creates vertical columns
- Holds actual content
- Syntax

```
<div class="col">
```

W-100

The Bootstrap utility class **.w-100** is used to set the width of an element to 100% of its parent element's width.

Example

```
<!DOCTYPE html>

<html lang="en">

<head>

   <meta charset="UTF-8">

   <meta name="viewport" content="width=device-width, initial-scale=1.0, shrink-to -fit=no" >

   <title>First Bootstrap Program</title>

   <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.8/dist/css/bootstrap.min.css" rel="stylesheet" integrity="sha384-sRIl4kxILFvY47J16cr9ZwB07vP4J8+LH7qKQnuqkuIAvNWLzeN8tE5YBujZqJLB" crossorigin="anonymous">

   <link rel ="stylesheet" href="style.css">
</head>
<body>
   <div class="container">
       <div class="row">
```

```
            <div class="col">col1 </div>
             <div class="col">col2 </div>
             <div class="col">w-100"></div>
             <div class="col">col3 </div>
             <div class="col">col4 </div>
             <div class="col">col5 </div>
             <div class="col">col6 </div>
       </div>
 <div class="row">row 2</div>
  <div class="row">row 3</div>
   </div>
</body>
</html>
```

Style.css

```
.container{

   border:8px solid black;
}
.row{

   border:5px solid red;
}
.col{

   border:5px solid greenyellow;
}
```

<u>.row-cols-2</u>

.row-cols-2 is a Bootstrap grid utility class that automatically creates 2 equal-width columns per row inside a .row.

Example

```
<!DOCTYPE html>
<html lang="en">
<head>
   <meta charset="UTF-8">
   <meta name="viewport" content="width=device-width, initial-scale=1.0, shrink-to -
  fit=no" >
   <title>First Bootstrap Program</title>
   <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.8/dist/css/bootstrap.min.css"
  rel="stylesheet" integrity="sha384-
  sRIl4kxILFvY47J16cr9ZwB07vP4J8+LH7qKQnuqkuIAvNWLzeN8tE5YBujZqJLB
  " crossorigin="anonymous">
   <link rel ="stylesheet" href="style.css">
</head>
<body>
```

```
  <div class="container">
      <div class="row row-cols-3">
          <div class="col">col1 </div>
           <div class="col">col2 </div>
           <div class="col">col3 </div>
           <div class="col">col4 </div>
           <div class="col">col5 </div>
           <div class="col">col6 </div>
     </div>
 <div class="row">row 2</div>
  <div class="row">row 3</div>
   </div>
</body>
```

Why Learn Bootstrap?

- Accelerates the development of modern web pages.
- Ensures consistent design across devices and browsers.
- Reduces the need for writing custom CSS.
- Ideal for both beginners and professionals in web development.

Bootstrap Grid System

The Bootstrap Grid System is a responsive layout system that helps developers create web page layouts using rows and columns.
It is based on Flexbox and divides the page into 12 equal columns.

It helps to:

- Create responsive designs
- Align content easily
- Arrange elements in rows and columns
- **Example:**

```
<!DOCTYPE html>
<html lang="en">
<head>
   <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1, shrink-to-fit=no" >
   <title>First Bootstrap Program</title>
   <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.8/dist/css/bootstrap.min.css" rel="stylesheet" integrity="sha384-sRIl4kxILFvY47J16cr9ZwB07vP4J8+LH7qKQnuqkuIAvNWLzeN8tE5YBujZqJLB" crossorigin="anonymous">
   <link rel ="stylesheet" href="style.css">
</head>
<body>
```

```
    <div class="container">
    <div class="row">
        <div class="col-12" id="Header">
        <h1>Website Name</h1>
        </div>
    </div>
 <div class="row">
<div class="col-12" id="Menu">
   <ul>
                <li><a href="#">Home</a></li>
                <li>|</li>
                <li><a href="#">About Us</a></li>
                <li>|</li>
                <li><a href="#">Gallery</a></li>
                <li>|</li>
                <li><a href="#">Contact Us</a></li>
            </ul>
</div>
 </div>
 <div class="row">
<div class="col-8" id="Content">
<h2>Sub Heading</h2>
<p>Lorem ipsum dolor sit amet consectetur, adipisicing elit. Id nisi numquam
  repellendus velit vitae? Consectetur numquam distinctio expedita, earum, ad neque
  deserunt excepturi soluta laudantium, laborum voluptatum nobis at ipsam?</p>
</div>
<div class="col-4" id="Sidebar">
<ul>
    <li><a href="">Home </li>
    <li><a href="">About Us</a></li>
    <li><a href="">Gallary</a></li>
    <li><a href="">Contact Us</a></li>
</ul>
</div>
</div>
<div class="row">
<div class="col-12" id="Footer">
 @Copy right
</div>
</div>
  </div>
</body>
</html>
```

Style.css

```
body{
background: grey;

}
#Header
{
background:orange;
padding:10px 0 10px 10px;
}
#Menu
{
background:pink;
padding:3px;
}
#Menu ul
{
list-style: none;
margin:0;
}
#Menu ul li
{
display: inline-block;
}
#Content
{
background:white;
}
#Sidebar
{
background:lightblue;

}
#Footer
{
background:lightblue;

}
```

Bootstrap classes

Bootstrap classes are predefined CSS classes provided by the Bootstrap framework to quickly design responsive and attractive web pages without writing custom CSS.

Example

1. Layout Classes (Grid System)

Used to create page structure and responsiveness.

1.Container

Class	Use
.container	Fixed width
.container-fluid	Full width
.container-sm/md/lg/xl	Responsive width

2. Row

Creates horizontal groups of columns.
<div class="row"></div>

3.Column

Divide space into 12 parts.

<div class="col-4">Column 1</div>

<div class="col-8">Column 2</div>

2. Typography Classes

Style text easily.

Class	Purpose
.h1 – .h6	Heading style
.lead	Bigger paragraph
.text-center	Center text
.text-primary	Blue text
.fw-bold	Bold
.fst-italic	Italic

3.Color Classes

For text and background colors.

Text colors
.text-primary
.text-success
.text-danger
.text-warning
.text-info
Background colors
.bg-primary

.bg-success
.bg-danger

4. **Spacing Classes (Margin & Padding)**

Very important for layout spacing.

Format:

m = margin

p = padding

t = top

b = bottom

s = start

e = end

x = left & right

y = top & bottom

0–5 = size

5.Button Classes

Create styled buttons.

Class	**Style**
.btn	Base
.btn-primary	Blue
.btn-success	Green
.btn-danger	Red
.btn-outline-*	Outline

6. **Display & Visibility Classes**

Class	**Purpose**
.d-none	Hide
.d-block	Block
.d-flex	Flexbox
.d-grid	Grid
.visible	Show

7.Flexbox Classes

Align items easily.

Class	**Use**
.d-flex	enable flex
. justify-content-center	horizontal center
.align-items-center	vertical center
. flex-column	vertical layout

8.Component Classes

Bootstrap ready-made UI components:

Component	Classes
Navbar	**.navbar**
Card	.card
Alert	.alert
Modal	.modal
Table	.table
Form	.form-control

Example

```
<!DOCTYPE html>
<html lang="en">
<head>
   <meta charset="UTF-8">
   <meta name="viewport" content="width=device-width, initial-scale=1.0">
   <title>Bootstrap All Classes Demo</title>

   <!-- Bootstrap CSS -->
   <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/css/bootstrap.min.css"
  rel="stylesheet">
</head>

<body class="bg-light">

<!-- ================= NAVBAR ================= -->
<nav class="navbar navbar-expand-lg navbar-dark bg-primary px-4">
   <a class="navbar-brand fw-bold" href="#">Student Portal</a>

   <button class="navbar-toggler" data-bs-toggle="collapse" data-bs-target="#menu">
      <span class="navbar-toggler-icon"></span>
   </button>

   <div class="collapse navbar-collapse" id="menu">
      <ul class="navbar-nav ms-auto">
         <li class="nav-item"><a class="nav-link active" href="#">Home</a></li>
         <li class="nav-item"><a class="nav-link" href="#">Courses</a></li>
         <li class="nav-item"><a class="nav-link" href="#">Contact</a></li>
      </ul>
   </div>
</nav>
<!-- ================= CONTAINER ================= -->
<div class="container mt-5">

   <!-- ================= ALERT ================= -->
   <div class="alert alert-success text-center">
```

```html
    Welcome to Bootstrap Dashboard 🚀
</div>
<!-- ================= TYPOGRAPHY ================= -->
<h1 class="text-center text-primary mb-4">Student Dashboard</h1>
<p class="lead text-center fst-italic">
    This project demonstrates all major Bootstrap classes
</p>
<!-- ================= GRID + CARDS ================= -->
<div class="row g-4 mt-3">

    <div class="col-md-4">
        <div class="card shadow p-3 text-center">
            <h5 class="card-title">Total Students</h5>
            <h2 class="text-success">150</h2>
            <button class="btn btn-success btn-sm">View</button>
        </div>
    </div>

    <div class="col-md-4">
        <div class="card shadow p-3 text-center">
            <h5 class="card-title">Courses</h5>
            <h2 class="text-info">12</h2>
            <button class="btn btn-info btn-sm">View</button>
        </div>
    </div>

    <div class="col-md-4">
        <div class="card shadow p-3 text-center">
            <h5 class="card-title">Attendance</h5>
            <h2 class="text-danger">85%</h2>
            <button class="btn btn-danger btn-sm">View</button>
        </div>
    </div>

</div>
<!-- ================= FLEXBOX ================= -->
<div class="d-flex justify-content-between align-items-center mt-5">
    <h3 class="text-secondary">Student List</h3>
    <button class="btn btn-primary">Add Student</button>
</div>
<!-- ================= TABLE ================= -->
<table class="table table-striped table-bordered table-hover mt-3 text-center">
    <thead class="table-dark">
        <tr>
            <th>ID</th>
            <th>Name</th>
```

```html
      <th>Course</th>
      <th>Status</th>
    </tr>
  </thead>

  <tbody>
    <tr>
      <td>1</td>
      <td>Rahul</td>
      <td>Java</td>
      <td><span class="badge bg-success">Active</span></td>
    </tr>
    <tr>
      <td>2</td>
      <td>Priya</td>
      <td>Bootstrap</td>
      <td><span class="badge bg-warning">Pending</span></td>
    </tr>
  </tbody>
</table>


<!-- ================= FORM ================= -->
<div class="card p-4 mt-5 shadow">
  <h4 class="mb-3">Contact Form</h4>

  <form>
    <div class="mb-3">
      <label class="form-label">Name</label>
      <input type="text" class="form-control">
    </div>

    <div class="mb-3">
      <label class="form-label">Email</label>
      <input type="email" class="form-control">
    </div>

    <div class="mb-3">
      <label class="form-label">Message</label>
      <textarea class="form-control"></textarea>
    </div>

    <button class="btn btn-primary">Submit</button>
    <button class="btn btn-outline-secondary">Cancel</button>
  </form>
</div>
```

```
</div>
<!-- ================= FOOTER ================= -->
<footer class="bg-dark text-white text-center p-3 mt-5">
   <p class="mb-0">© 2026 Student Portal | Built with Bootstrap</p>
</footer>
<!-- Bootstrap JS -->
<script
  src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/js/bootstrap.bundle.min.js"></script>

</body>
</html>
```

Bootstrap Components

Bootstrap components are pre-styled UI elements like buttons, forms, navbars, and more, built with HTML, CSS, and JavaScript.

They provide a consistent and responsive design framework, facilitating rapid development of web applications with minimal customization required.

Common Bootstrap Components List

Bootstrap provides:

1. Buttons
2. Alerts
3. Cards
4. Navbar
5. Forms
6. Tables
7. Modal
8. Carousel
9. Dropdown
10. Pagination
11. Badges
12. Progress bar
13. List group

1.Button- Used to create styled buttons.

Classes

btn btn-primary

btn-success

btn-danger

btn-warning

btn-outline-primary

Example

```
<button class="btn btn-primary">Save</button>
<button class="btn btn-danger">Delete</button>
```

2. Alert- Used to show messages or notifications.

Classes

alert alert-success

alert-danger

alert-warning

alert-info

Example

```
<div class="alert alert-success">
  Data saved successfully!
</div>
```

3.Cards- Used to display content inside boxes & Very useful for products, profiles, blogs

Example

```
<div class="card" style="width: 18rem;">
  <img src="img.jpg" class="card-img-top">
  <div class="card-body">
    <h5 class="card-title">Title</h5>
    <p class="card-text">Some text here</p>
    <a href="#" class="btn btn-primary">Read More</a>
  </div>
```

```
</div>
```

4. Navbar(Navigation Bar)

Used for menus/links.

Example:

```
<nav class="navbar navbar-expand-lg navbar-dark bg-dark">
  <div class="container">
    <a class="navbar-brand" href="#">MySite</a>
    <div>
      <a class="nav-link text-white" href="#">Home</a>
      <a class="nav-link text-white" href="#">About</a>
    </div>
  </div>
</nav>
```

5.Forms- used to create input form

classes

form-control

form-label

form-check

form-select

Example:

```
<form>
  <label class="form-label">Email</label>
  <input type="email" class="form-control">

  <button class="btn btn-primary mt-2">Submit</button>
</form>
```

6.Tables- Used to style tables

Classes

table

table-striped

table-bordered

table-hover

Example:

```
<table class="table table-striped">
  <tr>
    <th>Name</th>
    <th>Age</th>
  </tr>
  <tr>
    <td>John</td>
    <td>20</td>
  </tr>
</table>
```

6.Modal (Popup Box)

Used to show popup windows.(Requires Bootstrap JS)

Example:

```
<button class="btn btn-primary" data-bs-toggle="modal" data-bs-target="#mymodal">
  Open Modal
</button>
<div class="modal fade" id="mymodal">
  <div class="modal-dialog">
    <div class="modal-content p-3">
      <h4>Hello Modal</h4>
    </div>
  </div>
</div>
```

7 **Carousel (Slider)**

Used for image sliders.

Example:

```
<div id="demo" class="carousel slide" data-bs-ride="carousel">
```

```
  <div class="carousel-inner">
    <div class="carousel-item active">
      <img src="img1.jpg" class="d-block w-100">
    </div>
    <div class="carousel-item">
      <img src="img2.jpg" class="d-block w-100">
    </div>
  </div>
</div>
```

8 Badges- Used for counts/labels.

Example:

```
<button class="btn btn-primary">
  Messages <span class="badge bg-danger">5</span>
</button>
```

9 Progress Bar- Shows progress status.

Example:

```
<div class="progress">
  <div class="progress-bar" style="width:70%">
    70%
  </div>
</div>
```

10 List Group-Used to display lists.

Example:

```
<ul class="list-group">
  <li class="list-group-item">Item 1</li>
  <li class="list-group-item">Item 2</li>
</ul>
```

Bootstrap as a cross platform

Bootstrap is considered a "cross-platform" framework because it allows developers to create responsive websites and applications that function consistently across various devices and browsers, including desktops, tablets, and smartphones, without needing to write significantly different code for each platform, thanks to its responsive design features and compatibility with major browsers.

1. **Work on Different Devices**

   - It Work on different devices like Mobile**,** Tablet**,** Laptop**,** Desktop
   - Because of Grid system, Responsive classes, Media queries

2. Work on Different Screen Sizes

   - Bootstrap uses **breakpoints**:

   col-sm

   col-md

   col-lg

   col-xl

So layout changes automatically.

3. **Work on Different Browsers**

   - Bootstrap supports:

✓ Chrome
✓ Firefox
✓ Edge
✓ Safari
✓ Opera

It uses:

- Standard CSS
- Normalized styles
- Cross-browser fixes

So the design looks same everywhere.

4. **Different Operating Systems**

It Works on Windows, Linux, macOS, Android,iO Because Bootstrap runs inside **web browsers**, not OS dependent.

Features of Bootstrap

- **Grid System:** Easily create responsive layouts with a flexible grid system that adjusts seamlessly to various screen sizes.
- **Forms**: Build user-friendly forms with diverse styles and functionalities, including validation.

- **Buttons**: Add customizable buttons of all shapes and sizes to enhance your website's interactivity.
- **Navigation:** Implement intuitive navigation menus, from simple dropdowns to advanced mega menus.
- **Alerts:** Effectively communicate with dismissible alerts, warnings, and success notifications.
- I**mages:** Improve your website's visual appeal with responsive image handling.
- JavaScript Plugins: Integrate interactive features like modals, tooltips, and carousels using Bootstrap's JavaScript plugins.

<u>Applications of Bootstrap</u>

- **Creating Responsive Websites**- Bootstrap is used to create websites that automatically adjust for: Mobile, Tablet, Laptop, Desktop
- **Web Application Interfaces**- Used to design **UI (User Interface)** of:
Login pages, Dashboards, Admin panels, Forms
- **Portfolio Websites:** Bootstrap helps students and professionals create:
Personal portfolio, Resume websites, Profile pages
- **E-commerce Websites:** Used to design: Product pages, Shopping carts
Checkout forms, Product cards
- **Landing Pages**: Bootstrap is useful for: Marketing pages, Product launch pages, Advertisement pages
- **Form And Data Collections:** Used for: Registration forms, Feedback forms, Contact forms, Surveys.

<u>W3C</u>

- W3C stands for World Wide Web Consortium.
- It is an international organization that develops standards and guidelines for the World Wide Web so that websites work properly on all browsers and devices.

<u>The World Wide Web Consortium (W3C) supports web technologies by</u>

1. Developing Web Standards

W3C creates official standards (specifications) for web languages.

Examples:

- HTML → webpage structure
- CSS → styling
- SVG → graphics
- XML → data sharing

These standards ensure:

✓ Same look in all browsers

✓ Error-free coding
✓ Easy development

2.Publishing Specifications

W3C publishes technical documents that explain:

- Syntax rules
- Features
- Usage methods
- Best practices

These are called Specifications.

Example:
HTML5 Specification explains:

- tags
- elements
- attributes
- APIs

3. Browser Compatibility

W3C makes sure that:

✓ Chrome
✓ Firefox
✓ Edge
✓ Safari

follow the same rules.

So websites behave the same everywhere.

4.Working Groups & Collaboration

W3C forms Working Groups of:

- Developers
- Companies
- Researchers
- Browser vendors

They:

- discuss problems
- design new features

- improve technologies

Example:
HTML Working Group
CSS Working Group

5. Testing & Validation Tools

W3C provides free tools to test web code.

Tools:

- HTML Validator
- CSS Validator
- Accessibility checker

These tools:
✔ Find errors
✔ Improve quality
✔ Follow standards

6. Accessibility Support (WCAG)

W3C supports people with disabilities.

They created:
WCAG (Web Content Accessibility Guidelines)

This ensures:
✔ Screen reader support
✔ Keyboard navigation
✔ Proper contrast
✔ Inclusive websites

7.Security & Privacy Standards

W3C helps improve:

✔ HTTPS
✔ Secure APIs
✔ Safe data handling
✔ User privacy

This protects users from cyber threats.

8 Continuous Updates

Web technology changes fast.

W3C:

- updates standards regularly

- adds new features
- removes outdated ones

Example:
HTML → HTML5 → HTML Living Standard

Advantages

- W3C enables the easier maintenance of the W3C validated websites.
- It provides a consistent and defined look for all the W3C validated websites.
- It standardizes the validated websites so that they are accessible to different devices.
- It enables faster browser interaction.

Disadvantages

- W3C validation is a timely process and thus the time for full validation depends on the website code.
- W3C validation exercises have costs associated with them.
- Sometimes translation issues arise in W3C validation of websites.